

Mixing = modulation

- A low pass signal (described by low frequencies) such as voice and video cannot be sent from New York to Seattle.
- In fact, voice loses power $\sim \frac{1}{d^2}$ (you cannot hear me speaking if you are 30 meters away)
- The fact that voice dissipates indicates that the environment (air) is resistant to low frequencies.
- We can transmit voice as is, but we will need an antenna of size 10-100 km.



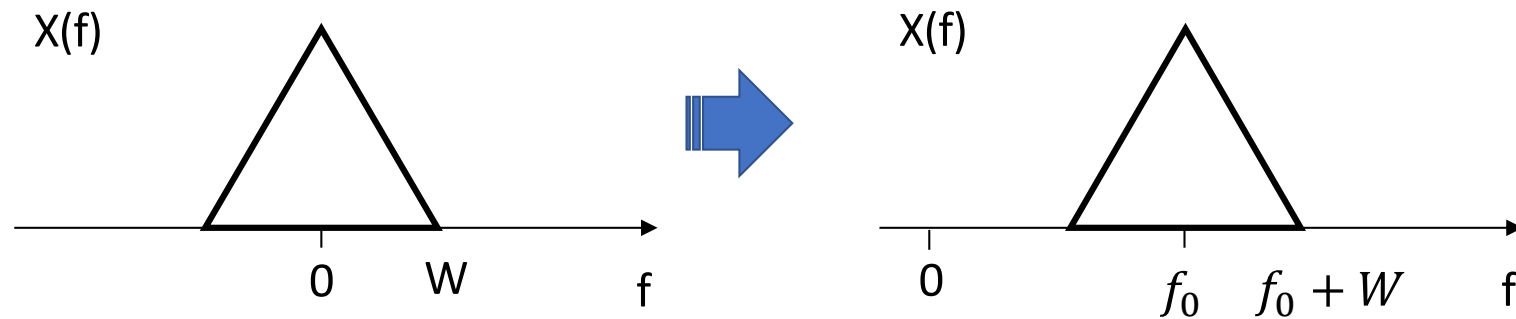
How to overcome the resistance of the air?

- Air (channel) acts as a band pass filter.
- Voice, video, etc. are low pass signals.
- To pass voice over channel, we need to translate voice to medium frequency.
- This is done through modulation (mixing).



How do we do this?

We want:



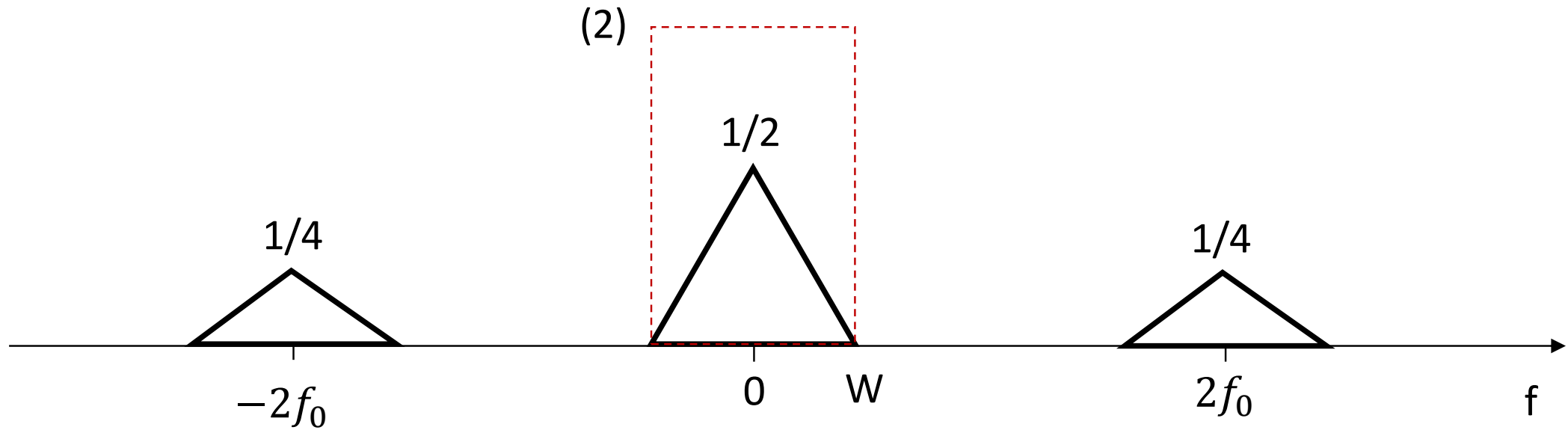
Mathematically it is described as:

$$x_m(t) = x(t) \times \cos(2\pi f_0 t)$$

By duality property, the FT of the modulated signal is

$$X_m(f) = \frac{1}{2} X(f \pm f_0)$$

Passes through environment!

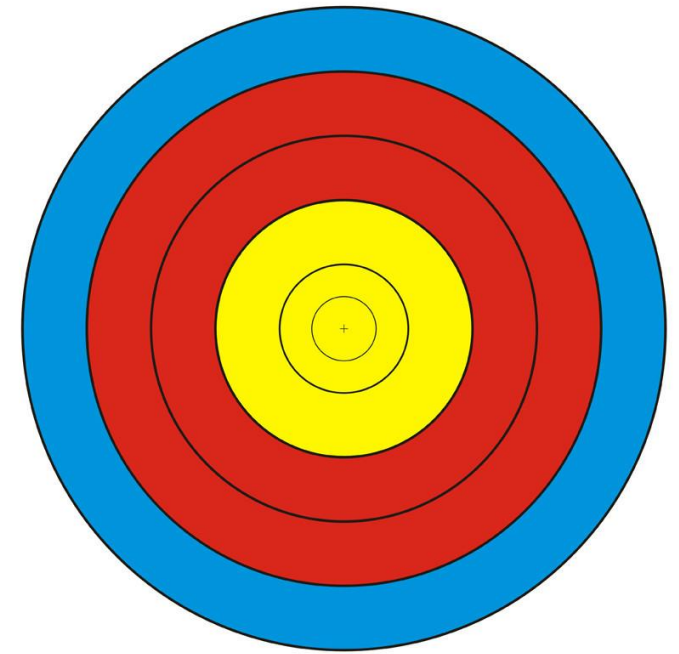
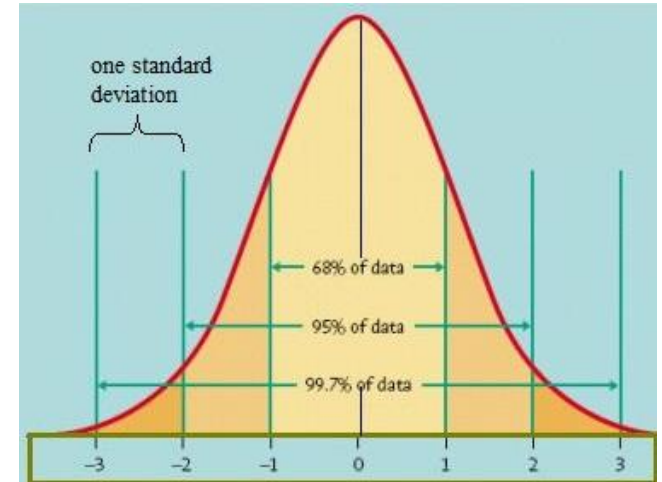


Demodulation (how to extract voice from modulated signal?)

- Multiply the modulated signal by the same cosine function.
- $x_m(t) \times \cos(2\pi f_0 t)$
- Pass the result through a low pass filter (set $f_{cutoff} = W$)
- Amplify the signal

A random experiment

- Random experiment = outcome cannot be predicted
- Example: Bow and arrows (2d target)
- Outcomes and their probabilities = weights (how often do we observe an outcome?)
- Fit a 2D function into the point on the target and look at the slice – it is a Gaussian bell.



Gaussian probability density function

- The bell is mathematically described as

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

m and σ^2 are mean and variance of this random variable.

- Noise due to a receiver (telescope or antenna) is Gaussian.
- Its mean is zero.
- Its variance can be estimated by using multiple measurements of the noise:

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - m)^2$$

where

$$m = \frac{1}{N} \sum_{i=1}^N x_i$$